Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

Bab 1

PENDAHULUAN

The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

Bab 2

TINJAUAN PUSTAKA

Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Daftar Isi

1 PENDAHULUAN **1**

2 TINJAUAN PUSTAKA **2**

3 Introduction **1**

4 Literature Review **2**
4.1 section1 2
4.1.1 title 2 2

5 System Development **4**
5.1 section1 4
5.1.1 title 2 4

6 Results and Discussion **5**
6.1 section1 5
6.1.1 title 2 5

7 Conclusion **7**

Daftar Gambar

4.1 Autonomous System Hierarchy 2
4.2 Sine Wave 3
4.3 Cosine Wave 3
4.4 The Sine and Cosine waves 3

Daftar Tabel

6.1 test table 6

Bab 3

Introduction

[1]

Bab 4

Literature Review

Each chapter is to begin with a brief introduction (in 4 or 5 sentences) about its contents. The contents can then be presented below organised into sections and subsections.

Technical writing is writing or drafting technical communication used in technical and occupational fields[1], such as computer hardware and software[2], engineering, chemistry, aeronautics, robotics, finance[3], medical, consumer electronics, biotechnology, and forestry. Technical writing encompasses the largest sub-field in technical communication. See figure 4.1 that shows the autonomous systems in Internet.

Gambar 4.1: Autonomous System Hierarchy

4.1 section1

[2]

4.1.1 title 2

[3]
The system is described by the equation 4.1 below. Here y is the ordinate and x is the abscissa , m is the slope and c a constant.

$$y = mx + c \tag{4.1}$$

Page centered and unnumbered multiple equations. The * symbol supresses equation numbering.
2x - 5y = 8
3x + 9y = -12
Side by side figures can be created using this environment. See fig 4.4 below.

[b]0.4

Gambar 4.2: Sine Wave

[b]0.4

Gambar 4.3: Cosine Wave

Gambar 4.4: The Sine and Cosine waves

Bab 5

System Development

Each chapter is to begin with a brief introduction (in 4 or 5 sentences) about its contents. The contents can then be presented below organised into sections and subsections.

5.1 section1

[2]

5.1.1 title 2

[3]

Bab 6

Results and Discussion

Each chapter is to begin with a brief introduction (in 4 or 5 sentences) about its contents. The contents can then be presented below organised into sections and subsections.

6.1 section1

[2]

6.1.1 title 2

[3]

Tabel 6.1: test table

Sl. No	Item 1	Itm 2
1	37	45
2	42	23
3	47	1
4	52	-21
5	57	-43
6	62	-65
7	67	-87
8	72	-109
9	77	-131
10	82	-153

Bab 7

Conclusion

Each chapter is to begin with a brief introduction (in 4 or 5 sentences) about its contents. The contents can then be presented below organised into sections and subsections.
[2]

References

[1] HU, Yun Chao, et al., *Mobile edge computing?A key technology towards 5G*, ETSI white paper, 2015, vol. 11, no 11, p. 1-16.

[2] @online Raspberry pi, `https://www.raspberrypi.org/` Online; accessed 10-June-2019

[3] HU, Yun Chao, et al., *Mobile edge computing?A key technology towards 5G*, ETSI white paper, 2015, vol. 11, no 11, p. 1-16.